Russ Mukhtar

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PROFESSIONAL SUMMARY

AI product engineer and M.S. candidate in Artificial Intelligence and Data Science at San Francisco State University. Skilled in building and iterating human-AI interfaces that combine usability, experimentation, and technical depth. Experienced in rapid prototyping, full-stack development, and model integration using PyTorch, TensorFlow, and modern web frameworks. Passionate about creating intuitive products that make AI collaboration natural and empowering.

SKILLS

Core: Product Prototyping | Full-Stack Development | Human-AI Interaction | UX Design | Rapid Iteration | AI Integration
AI & ML: Deep Learning | PyTorch | TensorFlow | Computer Vision | OpenCV | YOLO | Transfer Learning
Programming: Python | JavaScript | TypeScript | C++ | SQL | HTML/CSS | Bash | Git | Docker | Linux
Tools: Flask | React | Node.js | Next.js | Tailwind | MATLAB | Gazebo | Ansys | PyQt5 | Raspberry Pi
Languages: English (Fluent) | Russian (Fluent) | Turkish (Fluent) | Azerbaijani (Fluent) | Spanish (Beginner)

WORK EXPERIENCE

San Francisco State University — **San Francisco, CA**
Teaching Assistant — 08/2024 – Present

- Led Python installation and tutorial sessions for Theory of Computing students, ensuring all attendees successfully set up their development environments, improving their readiness for advanced coursework
- Provided individualized support to students during lab sessions, clarifying complex theoretical concepts and troubleshooting technical issues, resulting in higher student engagement and comprehension
- Collaborated with Professor to develop teaching materials and assist with grading, contributing to efficient course management and enhanced learning outcomes for over 30 students

RevSolz Corp. — **Calgary, AB Canada**
AI Team Lead — 07/2020 – 07/2023

- Led a team of data science interns to analyze oil industry data using machine learning models, which improved operational efficiency identification with 98% accuracy, enhancing decision-making processes company-wide
- Developed and optimized end-to-end machine learning pipelines for data preprocessing, model training, and testing, which increased model accuracy and reduced deployment time by 20%, improving production readiness
- Mentored and guided interns in machine learning techniques while maintaining clear communication with project managers, resulting in improved team productivity and two interns being offered part-time positions for their exceptional contributions

EDUCATION

San Francisco State University — **San Francisco, CA**
Master of Science in Artificial Intelligence and Data Science — Expected 05/2026

Graduate Thesis
AI-Assisted Medial-Axis Modeling of the Human Fallopian Tube — 08/2025 - Present
Mentor: Prof. Kazunori Okada, San Francisco State University

- Developing a 3D deep learning pipeline for automated segmentation and morphometric modeling of the human fallopian tube from micro-CT scans.
- Building a 3D U-Net (PyTorch, MONAI) to achieve ≥0.85 Dice accuracy in lumen segmentation and applying medial-axis extraction to generate centerlines with radius, curvature, and torsion descriptors.
- Implementing a Python-based analysis toolkit (VTK, VMTK, SciPy) for quantitative shape descriptors and segment classification.
- Aims to create a reproducible, simulation-ready dataset that bridges medical imaging, AI automation, and biomechanical modeling.

Yildiz Technical University **Istanbul, Turkey**
Bachelor of Science in Mechatronics Engineering 06/2023

Research

Digital Holographic Microscope 08/2022 - 06/2023

- Led the design and development of a Digital Holographic Microscope using Raspberry Pi 4, improving image resolution and system performance, which contributed to more precise scientific analysis
- Developed an intuitive user interface with PyQt5, streamlining complex functionalities and boosting operational efficiency, enhancing the overall user experience
- Collaborated with a team of three, integrating extensive research and problem-solving techniques, resulting in a successful project that gained faculty recognition for its innovation

Projects

AI Chatbot Web App 05/2025 - Present

- Designed and developed a full-stack conversational AI web application focused on UI and interaction quality over model complexity.
- Built backend using Flask and integrated Google's Gemma model for dynamic responses.
- Developed custom front-end interface in HTML/CSS/JavaScript emphasizing conversational flow, clarity, and emotional tone feedback.
- Iterated based on user testing to refine conversation pacing and visual cues for smoother human-AI collaboration.

DriveShare App & Website 05/2025 - Present

- Co-created a prototype platform that enables homeowners to rent out driveways to drivers seeking affordable parking.
- Developed both the website and mobile app versions using Next.js, Tailwind, and a PostgreSQL backend.
- Designed and implemented user flows, interface prototypes, and Figma wireframes for both driver and homeowner dashboards.
- Conducted iterative usability testing and refined UX design based on real user feedback, emphasizing seamless booking experience and intuitive navigation.
- Integrated backend functionality for listings, payments, and location-based driveway search to support full product flow.

Alzheimer's Disease Classification 08/2024 - 12/2024

- Developed deep learning models (VGG16, Inception V3, ResNet50) to classify PET scans into Alzheimer's stages, achieving up to 98% accuracy
- Preprocessed and augmented medical imaging data to ensure balanced classes and improved model generalization while adhering to rigorous evaluation protocols
- Fine-tuned transfer learning models with hyperparameter optimization and implemented advanced techniques like gradient clipping and learning rate schedulers for efficient training

DroneX (UAV Team) 09/2022 - 06/2023

- Led a software and AI team developing object detection and avoidance systems, resulting in securing First Place in the International Unmanned Aerial Vehicles competition
- Implemented YOLO v4 Tiny for real-time drone detection and spearheaded field data collection, significantly improving detection accuracy
- Programmed flight controllers using DroneKit and conducted Gazebo simulations on Linux, optimizing drone behavior and enhancing detection algorithms

Tennis Court Detection 02/2023 - 06/2023

- Developed Python-based algorithms for line, edge, and contour detection using OpenCV, accurately identifying tennis court lines in various conditions
- Built and curated a dataset of tennis court images and videos, facilitating the training and testing of detection models
- Optimized detection accuracy by applying edge detection methods and Hough Transform, fine-tuning algorithm parameters to handle noise and lighting variations